



CSE 4614: Technical Report Writing

Topics Covered in the $\text{\LaTeX}$ Learning Section

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May 18, 2026

Contents

1	Lab 1: Introduction to the Basics	1
2	Lab 2: Mathematical Expressions	2

1 Lab 1: Introduction to the Basics

The following topics are covered (not chronologically) in the $\LaTeX$ learning section:

- Structure of a $\LaTeX$ source code
 - The `documentclass` command
 - The preamble section
 - The document environment
 - Structure of a $\LaTeX$ command and environment
 - * General style of positional (mandatory) and optional arguments
- Inclusion of packages
 - Geometry package to change the paper type, margin, and layout of document
- Options for the `article` class to change font size and paper type
- Changing fonts by importing respective packages
- Use of `lipsum` package and command to generate filler text for type-setting
- Setting title, author, and date information and generating title with the `maketitle` command
 - Creating a separate title page with the `titlepage` environment
- Creating paragraphs and lines with line and paragraph breaks with the `\`, `newline` and `par` commands respectively
- Moving to new page with the help of `newpage` and `pagebreak` commands (there are slight differences, students can explore themselves)
- Making text bold, italic, and underlined
- Emphasizing text with the `emph` command and its behavior in the `article` class
- Single and multi-line comments (with the help of `comment` environment from the `verbatim` package)
- Changing text sizes, e.g. `large`, `Large`, `Huge`, `small`, `tiny`, etc.
- Creating numbered and unnumbered (asterisk `*`) version of commands) sections, subsections, and sub-subsections
- The `abstract` environment
- Centering text with the `center` environment (students should explore making text left or right aligned)
- Creating unordered (`itemize`) and ordered (`enumerate`) lists and nesting them
- Generating table of contents
- Hyperlinking items (basic) with the help of `hyperref` package

2 Lab 2: Mathematical Expressions

The following topics are covered (not chronologically) in the $\text{\LaTeX}$ learning section:

- The two mathematical modes/environments (how to and their differences):
 1. Inline mode, with:
 - `\(...\)`
 - `$...$`
 - `\begin{math}...\end{math}`
 2. Display mode, with:
 - `\[...\]`
 - `$$...$$`
 - `\begin{displaymath}...\end{displaymath}`
 - `\begin{equation}...\end{equation}` (*numbered*)
- The `amsmath` package and the `equation*` environment (all later `amsmath` environments have `*` variants)
- Difference between mathematical functions, operators, operands, and text in mathematical environments
- Mathematical commands, operators, symbols, such as `times`, super and subscripts, `frac`, `sin`, `cos`, `int`, `sum`, `prod`, `lim`, `limits` (to position lower and upper limits vertically)
- Equality, non-equality, greater than, less than, greater equal, less equal symbols
- `left` and `right` commands to match bracket pairs and `.` for open brackets
- Greek letters, `mathcal`, `mathfrac`, `mathbb` from `amssymb` (also show `multi-import` in a single `usepackage` command)
- Infinity (`\infty`), arrows, and spacing with `quad`, `qquad`, and likes of `\.`
- `\mathbf` and `\text` commands
- Referencing an equation, first with vanilla `ref` and then with the help of `autoref` command from `hyperref` package
- Vector symbol and center dot (`\cdot`) for dot product symbol
- Different `amsmath` environments and aligning equations:
 - `multiline`, `split` (inside another math environment), `align`, `gather`
- Different types of matrices with the help of `matrix`, `bmatrix`, `Bmatrix`, `pmatrix`, `vmatrix`, `Vmatrix` etc. environments